স্রষ্টা সহায়

Chapter-1.

Internet :
An internet is two or more networks that can communicate with each other.

LAN+WAN = ethernet

Host = end system ; network এর সাথে connected

Packet switches = routers , switches

communication links = fiber, copper, radio, satellite.

Networks = Connection of devices, routers, links.

Protocols = HTTP (web), skype, TCP, IP, wifi, 4G, Ethernet.

*All communication activity in Internet governed by protocol.

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Standards — RFC = Request for comments

IETF = Internet Engineering Task ~~Fre~~ Forse

* Infrastructure: provides services to applications

Network edge — hosts, access networks, physical media.

hosts = clients and servers

servers often in data centers

Hop — router থেকে router এর distance (device to device)

* How to connect end systems to edge router?

Ans: – residential access nets

– institutional access networks

– mobile access networks

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*Difference between Edge router and core router

Edge router	core router
1. Edge router characterize and safeguard IP traffic from other edge routers and core routers.	1. Core router performs centralized routing for subnetworks in business.
2. Edge router provide more protection on traffic compared to core router.	2. Core router provide less protection on traffic compared to edge router.
3. It is located at the edge.	3. It is located at the core.
4. It handles traffic from local users or devices.	4. It handles large volume of traffic between different parts of the network.
5. A failure at the network edge affects a smaller portion of network.	5. A failure in a core router can have a widespread impact on network connectivity.

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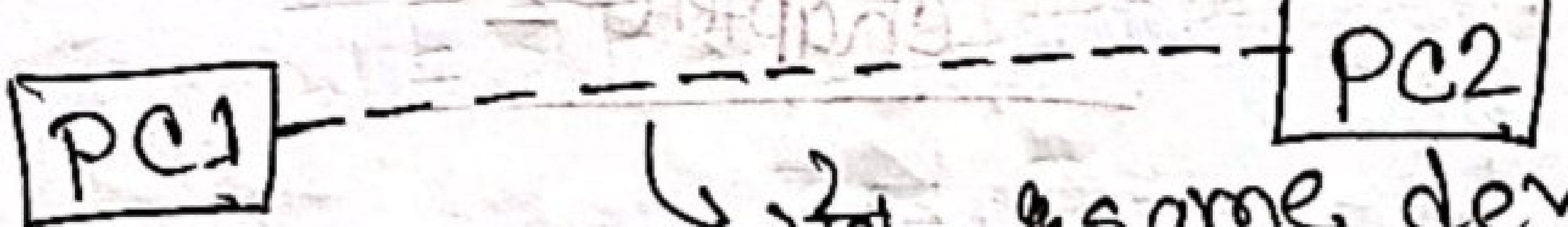
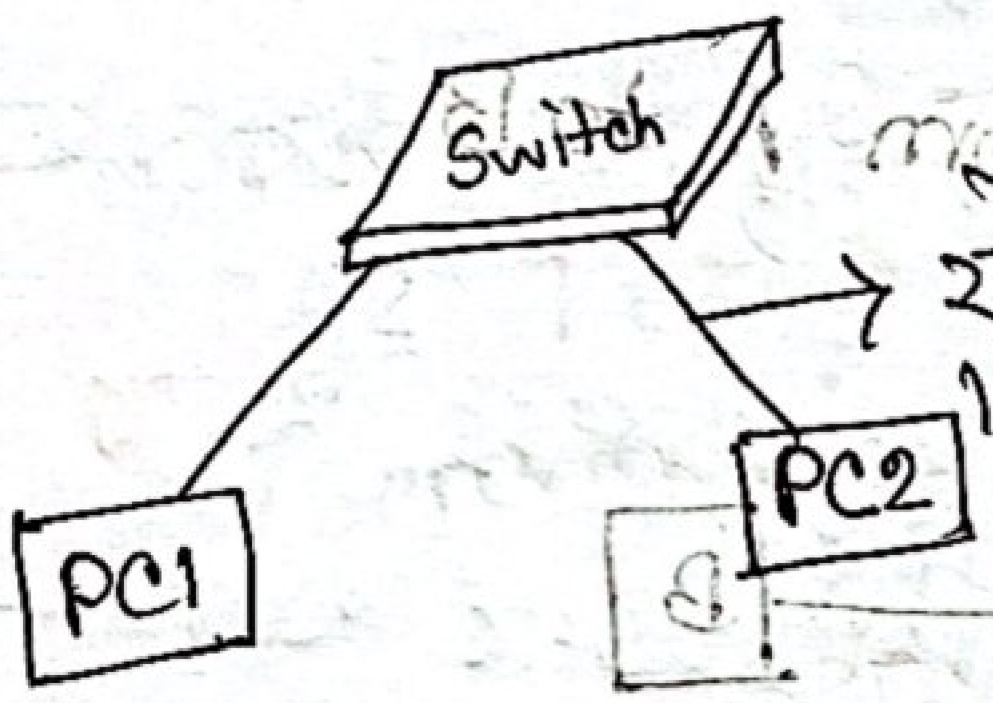
*Lab এর জন্য:



২টা same device এর জন্য cross cable use করতে হবে।



২টা different device এর জন্য straight line cable use করতে হবে।

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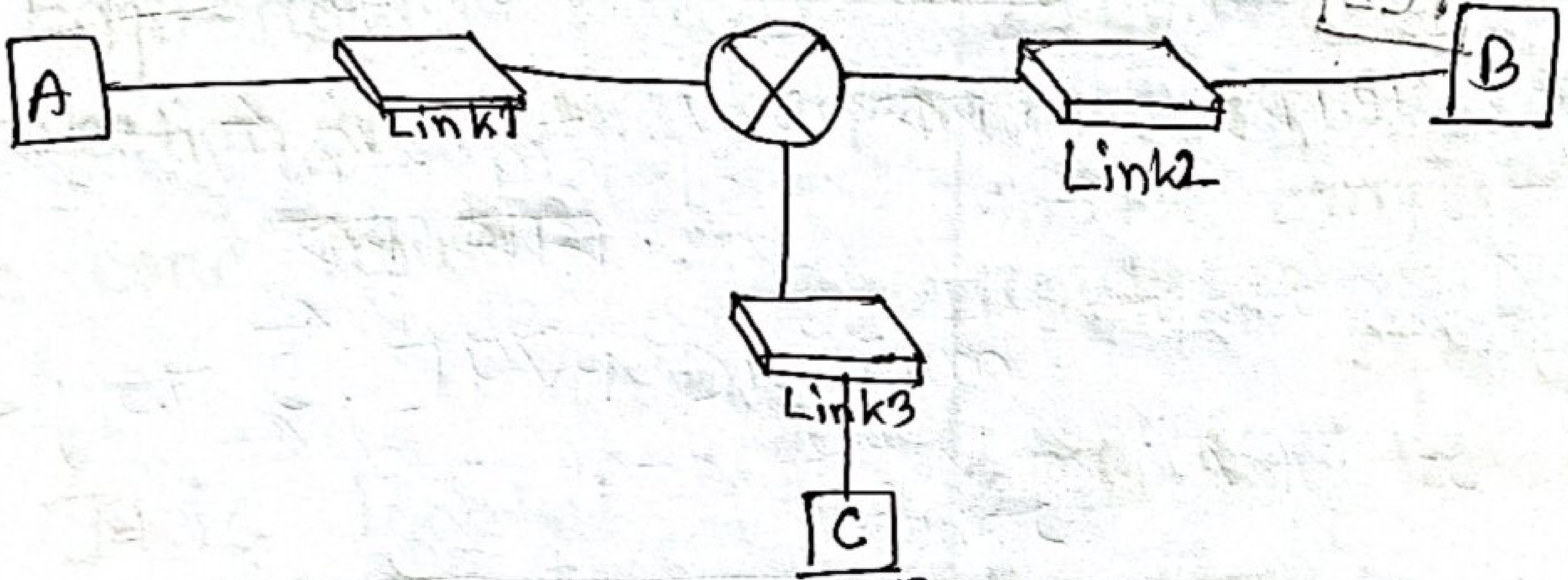
TCP/IP Protocol

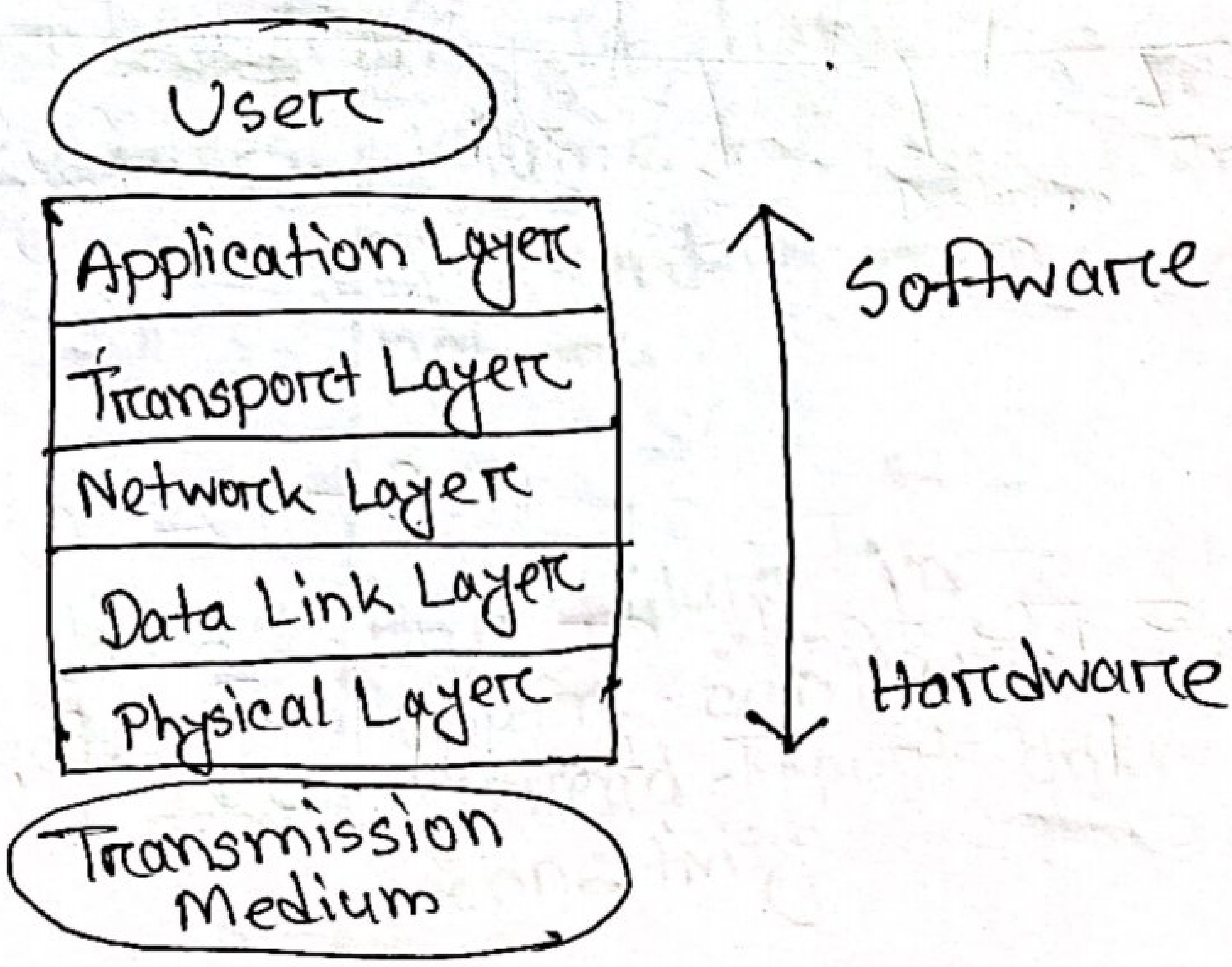
Chapter-2

Internet: An internet is two or more networks that can communicate with each other.

Communication from A to B



Internet Layer Model: The internet Protocol stack



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⇒ Application Layer:

Responsible for providing services to the user. This is the only layer to interact with user.

⇒ Transport Layer:

Responsible for delivery of a message from one process to another.

Duties/Services:

- → Port addressing
- → Segmentation and reassembly
- → Connection control
- → Flow control } End to end
- → Error control }

⇒ Network layer:

Responsible for the delivery of packets from the original source to the destination.

Duties/Services:

- → Logical addressing
- → Routing

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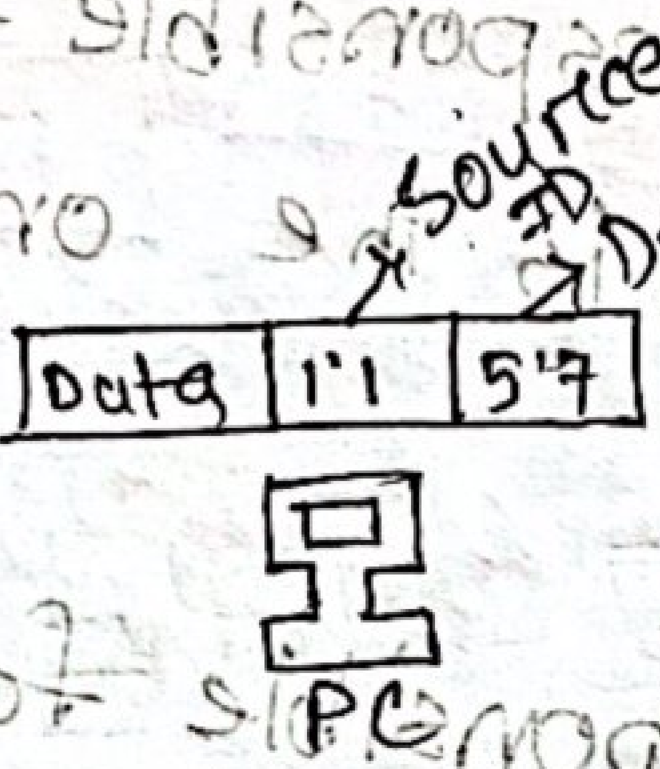
প্রত্যেক PC এর একটা unique id থাকে 32 bit এর (IP address)



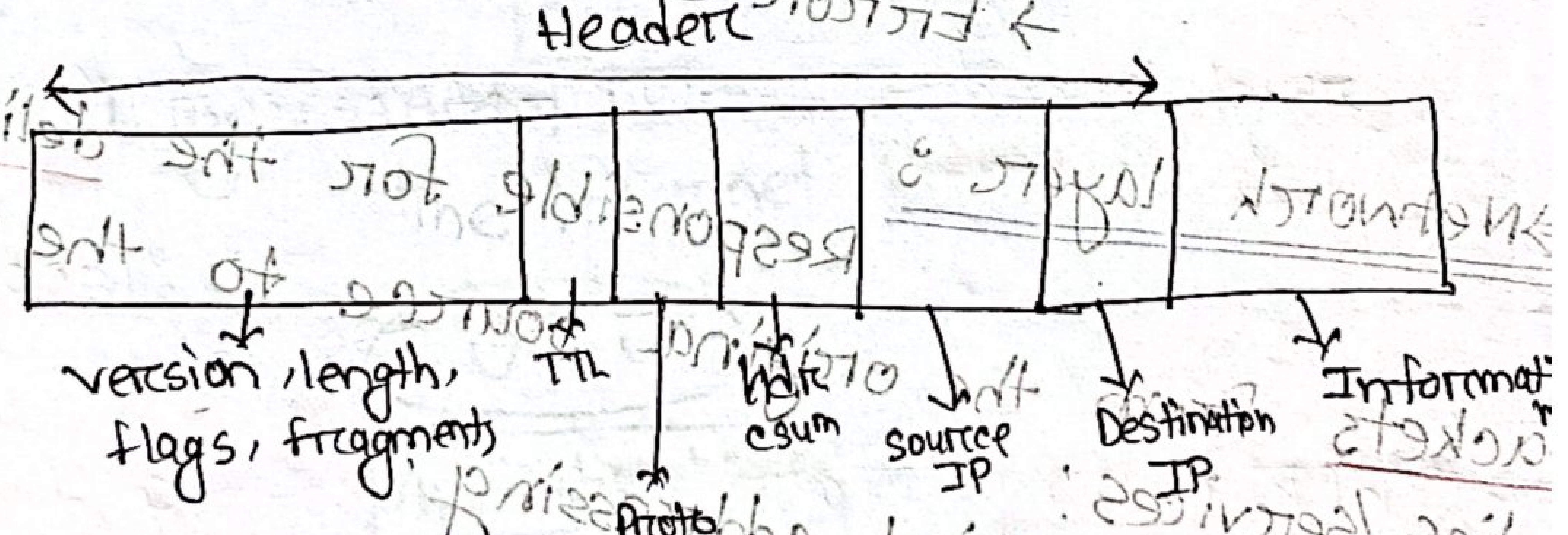
→ Routing algorithm use করা হয়

→ Routing → cost, traffic দেখে পথ ঠিক করবে

→ Hop-to-Hop (সেগুলো) mac address যাবে.

→ Subnet use করলে বেশি user connect করা যায়.

Layer 3: IPV4 Diagram



* Autonomous System — ASN

* Subnet ~~mask~~ এ 254 টা IP client এর জন্য use করা হয়। 00 & Last IP special

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Data Link Layer:

Responsible for transmitting frames from one node to the next.

Duties/services:

→ Framing
→ Physical addressing
→ Flow control } Hop-to-Hop
→ Error "
→ Access control

*Mac address/physical address → hexadecimal number
*Mac address change হয় hop-to-hop.

Physical Layer:

Responsible for transmitting individual bits from one node to the next.

Duties/service:

→ Physical characteristics of interfaces and media
→ Representation of bits.
→ Data rate (transmission rate)
→ Synchronization of bits.

*wireless হলে digital.

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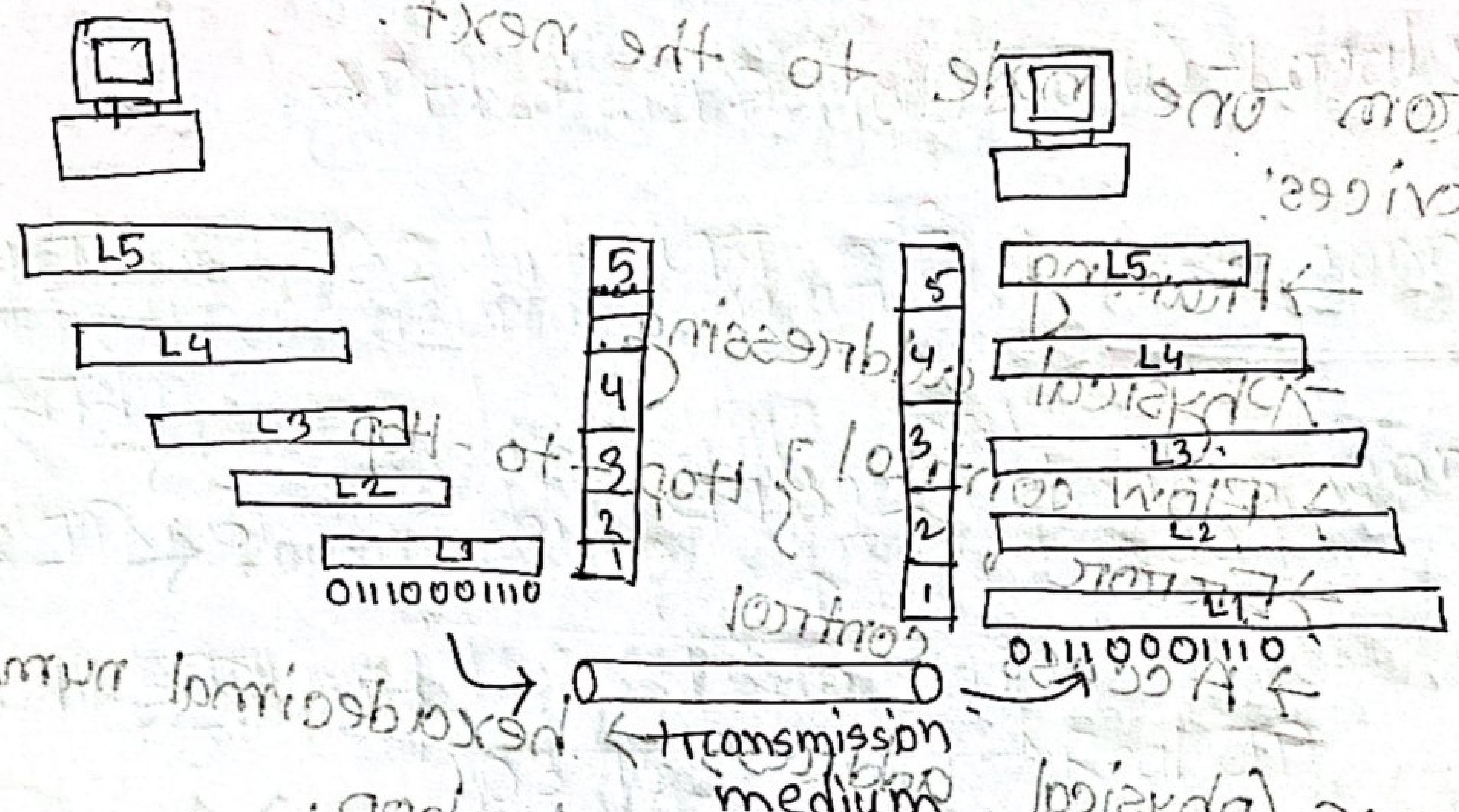
*The Big Picture:



*Internet model:

→ Node-to-node/Hop-to-Hop: Data Link layer

→ Host-to-host: Network layer

→ Process-to-process: Transport layer

*Protocol:

A set of rules

Internet (Protocol Suite (কোন layer এ কোন protocol)

→ Application = HTTP, FTP, Telnet, SMTP

→ Transport = TCP, UDP, SCTP

→ Network = IP (IPv4), IPv6, ICMP, IGMP

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→ Data Link = Ethernet, Wi-Fi, PPP

→ Physical = RS-232, DSL, 10Base-T

* OSI Model : Open System Interconnection

Two additional layers ⇒
→ Presentation layer
→ Session layer.

__Session Layer :__ Responsible for establishing, ~~mag~~managing and terminating connections between applications.

Duties/services:
→ Interaction management
= Simplex, half-duplex, full-duplex
→ Session recovery

__Presentation Layer:__ Responsible for handling differences in data representation to application.

Duties/services:
→ Data translation
→ Encryption.
→ Decrytion
→ Compression.

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* IP address 2 types:
→ Public IP/Real IP
→ Private IP (10.32.0.0) - Local এর জন্য

* IPv4 32 bit এর 2^{32} devices connect করা যায়।

* IPv6 128 bit এবং 2^{128} devices connect করা যায়।

103.133.254.1 → Real IP
103.133.254.23 → PSTU - এর IP

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Chapter—5
IPv4 addresses

*IP address — identifier, uniquely identifies devices connected to the Internet.

computer — 4 addresses.

→ Port address
→ Logical/IP address
→ Physical/mac address.

[NIC - Network Interface Card
এটাই interface]

IP Layer = Network layer

*IPv4 :

- 32 bits long : 4 byte → 4 octet বলে
- → addresses are unique and universal
- → address space 2^{32} or 4,294,967,296.
- → numbers in base 2, 16, 256
- → Permutation করে total কতগুলো combination এর IP address পাওয়া যায় :

* Address space: total no. of addresses used by the protocol.

* Dotted-decimal notation:

Binary: 10000000 00001011 00000011 00011111

Dotted decimal: $\boxed{128.11.3.31}$

128	64	32	16	8	4	2	1
7	6	5	4	3	2	1	0

* Positional value যোগ করে IP address বের করা।
* শুধু decimal এর ক্ষেত্রে (.) dot use করা হয়
* IP address এর range 0-255-র বেশি হওয়া যাবে না।

0.0.0.0
255.255.255.255 } reserved IP/special

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1st IP – Gateway হিসেবে

2nd IP – Special

zero – 0b – binary
0d – decimal
0x – hexadecimal
0c – octal

} এটা আমরা বসবে or base subscript হিসেবে।

* IP address base – 256

Example – 5.1

a) 10000001 . 00001011 . 00001011 . 11101111
129 . 11 . 11 . 239

replacing each group of 8 bits with its equivalent decimal number.

Example – 5.2

a) 111 . 56 . 45 . 78
01101111 00111000 00101101 01001110

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Example - 5.31

Find the errors, if any in the following IPv4 address.

a) 111.56.045.78

⇒ There should be no leading zeros in dotted decimal notation (045)

b) 221.34.7.8.20

⇒ We may not have more than 4 bytes in an IPv4 address.

c) 75.45.301.14

⇒ Each byte should be less than or equal to 255; 301 is outside this range.

d) 11100010.23.14.67

⇒ A mixture of binary and decimal notation is not allowed.

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Example - 5.4

a) 10000001   00001011   00001011   11101111

0X 81   0B   0B   EF

→ 0X is added at the begining to show that the number is in hexadecimal.

Example - 5.5 Find the number of addresses in a range if the first address is 146.102.29.0 and last address is 146.102.32.255

Solution:

1st address → 146.102.29.0

2nd " → 146.102.32.255

bit by bit subtract (base 256)

$$
\begin{array}{r}
146.102.29.0 \\
146.102.32.255 \\
\hline
0.0.3.255
\end{array}
$$

convert it to base 10 → $0\times256^3 + 0\times256^2 + 3\times256^1 + 255\times256^0 = 1023$

add 1 to the result →

Number of address $= 1023 + 1 = 1024$

Example-5.6

First address = 14.11.45.96

Number of address = 32

Number of address - 1 ⇒

32 - 1 = 31

base 256 ⇒ 0.0.0.31

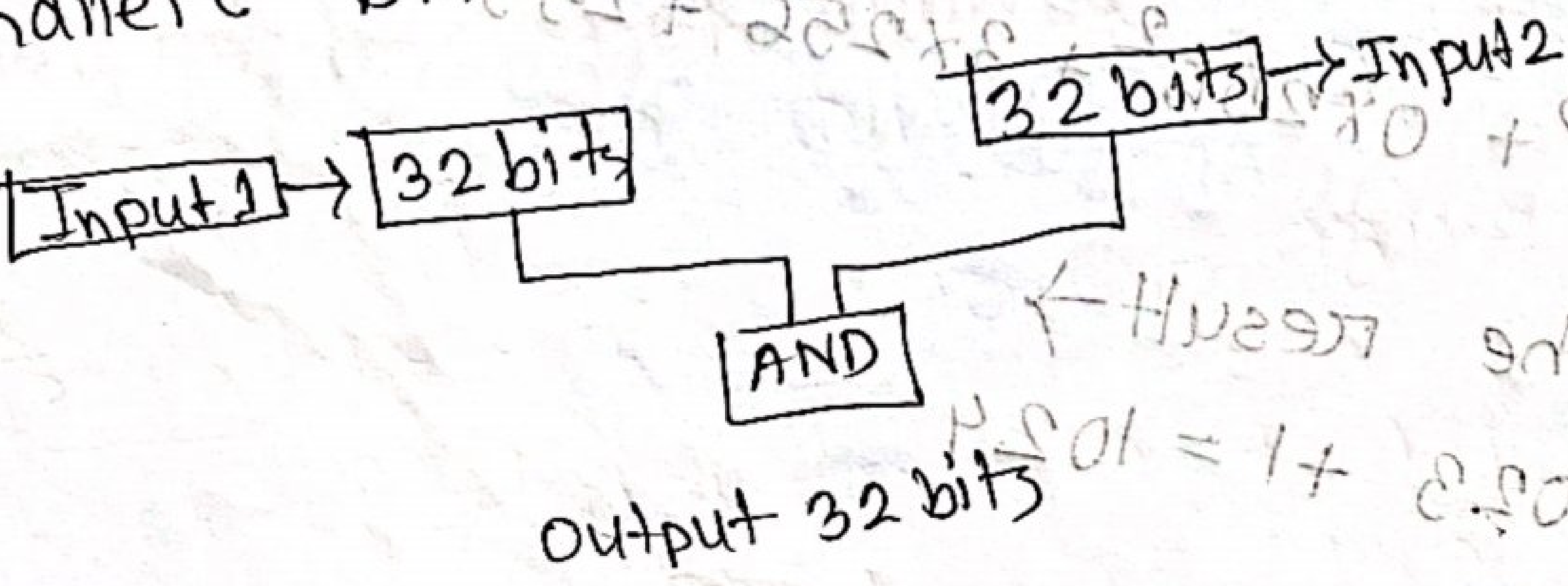
add it to the first address ⇒

Last address = $(14.11.45.96 + 0.0.0.31)_{256}$

= 14.11.45.127 (base 256)

*Bitwise AND operation:

The AND operation compares the two corresponding bits in two inputs and selects the smaller bit from the two.



AND

Input1	In2	out
0	0	0
0	1	0
1	0	0
1	1	1

~~There~~
When the numbers are represented in dotted decimal notation, we can use two shortcuts:

1. When at least one of the numbers is 0 or 255, the AND operation selects the smaller byte.
2. When none of the two bytes is either 0 or 255, we can write each byte as the sum of eight terms, where each term is a power of 2 or 2^n. We then select the smaller term in each pair and add them to get the result.

Example:- 5.8

First Num :	00010001	01111001	00001110	00100011
Second " :	11111111	11111111	10001100	00000000
Result :	00010001	01111001	00001100	00000000

we can solve using dotted decimal representation

First Num : 17 . 121 . 14 . 35

Second " : 255 . 255 . 140 . 0

Result : 17 . 121 . 12 . 0

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সেটা নিব

Powers	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Byte(14) :	0	0	0	0	8	4	2	0
Byte (140) :	128	0	0	0	8	4	0	0

Result (12) : 0 + 0 + 0 + 0 + 8 + 4 + 0 + 0 → 12

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সেটা নিব

same
তাই same
টাই নিব.

*Why ~~and~~ AND operation needs to be done?

~~operation is important be for:~~

Ans: AND operation

→ Subnetting : AND operation is used to compare the IP address of a device with its subnet mask to determine which subnet it belongs to.

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→ Routing: By performing a bitwise AND operation between the destination IP address of a packet and the router's routing table entries, the router can determine the appropriate outgoing interface or next hop for the packet to reach its destination.

→ Access control: By performing an AND operation between the packet's IP address and the mask specified in an ACL rule, the network device can decide whether to permit or deny packet's passage.

* Bitwise OR operation: The OR operation compares the corresponding bits in the two numbers and selects the larger bit from the two. Two short-cuts are:

1. When at least one of the two bytes is 0 or 255, the OR operation selects the larger byte.
2. When none of the two bytes is 0 or 255, we can write each bytes as the sum

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of eight terms is a power of 2, we then select term in each pair and add them to get the result of OR operation.

Example - 5.9

First num:	00010001	01111001	00001110	00100011
Second num:	11111111	11111111	10001100	00000000
Result	11111111	11111111	10001110	00100011

We can solve using dotted decimal representation:

First num:	17	121	14	35
Second num:	255	255	140	0
Result:	255	255	142	35

Power:	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Byte(14):	0	0	0	0	8	4	2	0
Byte(140):	128	0	0	0	8	4	0	0

Result: 128 + 0 + 0 + 0 + 8 + 4 + 2 + 0

→ 128+8+4+2 = 142

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* Two types addresses:
→ classful
→ classless

* Address space occupation:-

Octet1

→ Class A = 0 ; 1st address, 2^{31} (50%)

→ Class B = 10 ; 2^{30} (25%) [Future]

→ Class C = 110 ; 2^{29} (12.5%) [Multicast]

→ Class D = 1110 ; 2^{28} (6.25%)

→ class E = 1111 ; 2^{28} (6.25%)

Binary notation

* Dotted-decimal notation:-

→ Class A : 0 - 127

→ Class B : 128 - 191

→ Class C : 192 - 223

→ Class D : 224 - 239

→ Class E : 240 - 255

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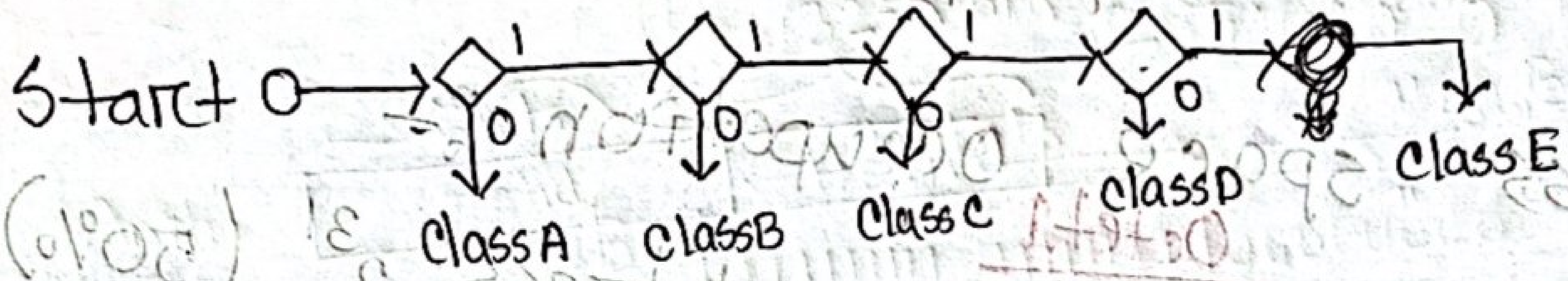
*Finding the address class using continuous checking:



Example - 5.10

a) 11000001 10000011 00011011 11111111

⇒ The first two bits are 1; the third bit is 0. This is a class C address.

b) 10100111 11011011 10001011 01101111

⇒ The first bit is 1; the second bit is 0. This is a class B address.

Example - 5.11

a) 227.12.14.87

⇒ The first byte is 227 (between 224 to 239) the class is D.

b) 193.14.56.22

⇒ The first byte is 193 (between 192 - 223) the class is C.

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*** Netid and Hostid:** → defines a particular host connected to that network.

Netid → defines the network

	Byte 1	Byte 2	Byte 3	Byte 4
Class A	Netid	Hostid (2^{24})		
" B	Netid		Hostid (2^{16})	
" C	Netid			Hostid (2^{8}) → বেশির use হয়েছে
" D	Multicast			
" E	Reserved for future use			

*** Two level of Addressing:**

netid	hostid
n bits	(32-n) bits

32 bits

class A ⇒ n = 8

" B ⇒ n = 16

" C ⇒ n = 24

D = 32

E = 40

*** Extracting information in a block:**

* 1st address → netid-এর পরের host portion 0 দি

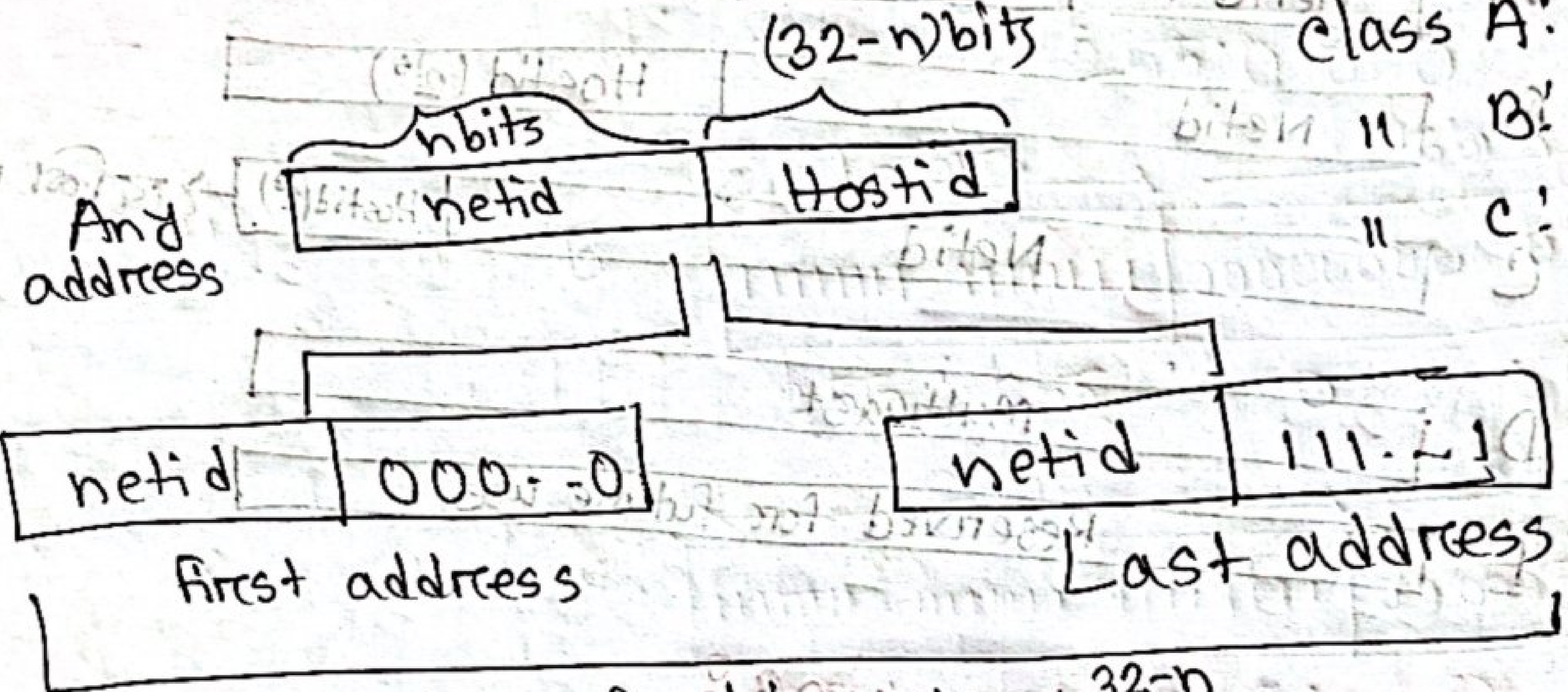
* Last " → " " " " 1 "

*** Total address space,** $N = 2^{32-n}$

⇒ To find the first address, we keep the n leftmost bits and set (32-n) rightmost bits to 0s.

⇒ To find last address, we keep n leftmost bits and set the (32−n) rightmost bits to 1s.

class A: $n=8$
" B: $n=16$
" C: $n=24$



Number of address: $N = 2^{32-n}$

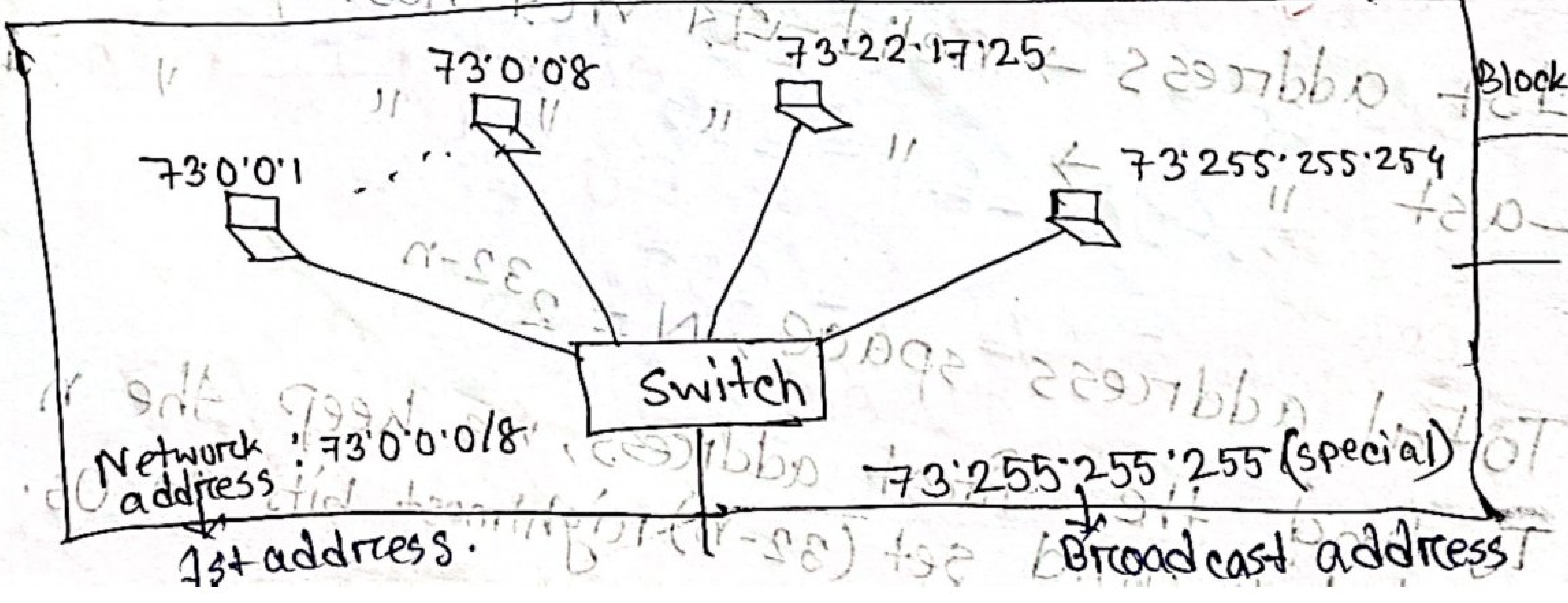
Fig: Information extraction in classful addressing

Example-5.13

73.22.17.25

73 is between 0−127. it is class A.

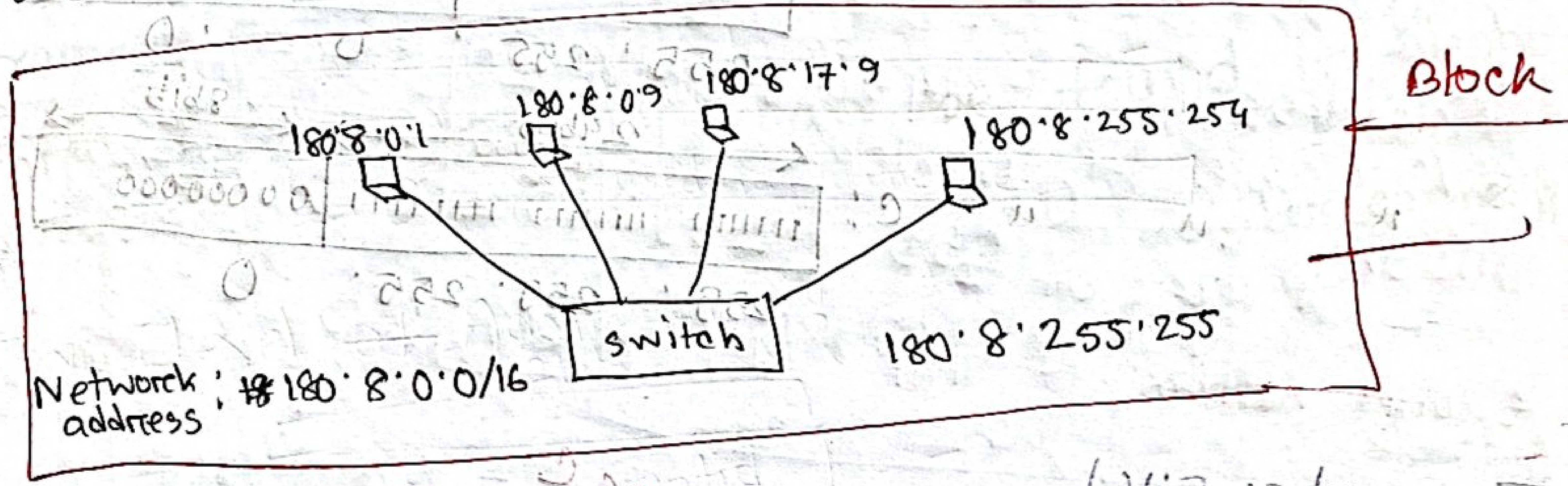
The value of $n = 8$



Example 5.14

180.8.17.9

180 is between 128 to 191. This class address is B. The value of n in B is 16.



Network address: Network address is also known as first address. It is used in routing a packet to its destination network. It is the identifier of a network.

Subnet Mask: Identify network address. Divides IP into network address and host address.

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*Network Mask:

Mask for class A:

← 8 bits →	← 24 bits →		
11111111	00000000	00000000	00000000
255	0	0	0

" " " B:

← 16 bits →		← 16 bits →	
11111111	11111111	00000000	00000000
255	255	0	0

" " " C:

← 24 bits →			← 8 bits →
11111111	11111111	11111111	00000000
255	255	255	0

Example:- 5.16)

class C

Destination address → $\boxed{201}$. 24 . 67 . 32

AND

Default mask → 255 . 255 . 255 . 0

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Network address → 201 . 24 . 67 . 0

*IP address wastage কমাতে - subnetwork করা হয়

* Three level addressing:

⇒ Subnetting :- Subnetting is a network that is divided into several smaller subnetworks with each subnetting having its own ~~set~~ subnetwork address.

Example - 5.17 :

Three level addressing can be found in telephone system:

~~626~~

(626) 358 - 1301 → subscriber connection.

area code, exchange

[/n → length of the netid]

⇒ when we divide a network into several subnetwork we need to create subnetwork mask.

⇒ when we divide a network into s number of subnetworks, each of equal numbers of hosts. then,

$$n_{sub} = n + \log_2 s$$

s → number of subnetwork.

n → length of ~~eo~~ netid

n_{sub} → length of each subnetid.

* $n_{super} = n - \log_2 s$

*Here, Network mask → default mask

Subnetwork address → ANDing operation.

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Example - 5.21 :

A network is divided into four subnets. One of the address in subnet 2 is 141.14.120.77. The subnet address is:

Address: 141 . 14 . 120 . 77 → $\log_2 2$ = ② → 2 टा 1 add

Mask : 255 . 255 . 192 . 0 (11000000)

Subnet Address: 141 . 14 . 64 . 0

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Address: (120) :	0 +	1 +	1 +	1 +	1 +	0 +	0 +	0
mask : (192) :	1 +	1 +	0 +	0 +	0 +	0 +	0 +	0
~~Resudtst~~ Result :	0 +	1 +	0 +	0 +	0 +	0 +	0 +	0

* Comparison between subnet mask and supernet mask:

Subnetting	Supernetting
1. It is the procedure to divide the network into subnetwork.	1. It is the procedure to combine ~~sub~~ small networks.
2. In subnetting network address is increased.	2. In supernetting Host address is increased.
3. The mask bits are moved towards right.	3. The mask bits are moved toward left.
4. It is implemented via variable length subnet masking.	4. It is implemented via classless interdomain routing.
5. It is used to reduce network congestion, improve security and optimize IP address allocation.	5. It is used to simplify routing, reduce routing table size, save RAM in routers.

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*Classless addressing :

Short term solution, IPv4

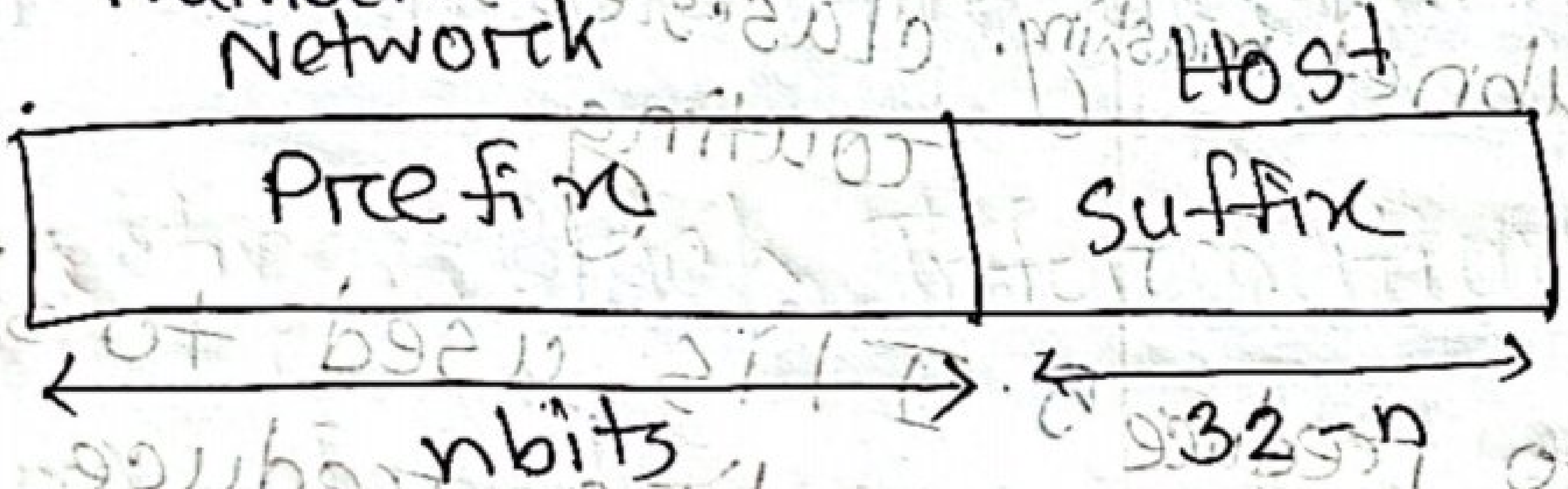
/n notation – prefix length + large block
– classless addressing ↳
use 23

N " – smal block

*Variable length block in Classless addressing.

Number of 1
↑
prefix – network address (n bits)

suffix – host " (32 – n bits)
↓
Number of 0



*The prefix length in classless addressing can be 1 to 32.

Example: 130.11.232.156/16
n = 16; prefix = 16
suffix = 16

Example 5.22 What is the prefix and suffix length if the whole internet is considered as one single block with 4294967296 addresses?

Solution: Prefix length 0

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suffix length 32:

All 32 bits vary to define $2^{32} = 4,294,967,296$ hosts in this single block.

~~Example 5.23~~ CIDR - classless Inter Domain Routing

* The only address in each block is defined by the block itself.

* Difference between classful address and classless address:

Classful address	Classless address
1. IP addresses are divided into five group using classful address	1. To prevent depletion of IP addresses classless address is used.
2. In classful addressing variable length subnet mask is not supported.	2. In classless addressing variable length subnet mask is supported.
3. It requires more bandwidth	3. It requires less bandwidth.
4. It does not import subnet mask.	4. It imports subnet mas
5. Address is divided into 3 parts: Network, Subnet, Host	5. Address is divided into two parts: Subnet and Host

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Classful address	classless address
6. Subnets are not displayed in other major subnet.	6. Subnets are displayed in other major subnet.
7. Faults can be detect easily.	7. Faults detection is little tough.
8. Classless Inter Domain Routing is not supported	8. Classless Inter Domain Routing is supported.

*In classless we need to know one of the addresses in the block and prefix length to define the block.

*Extracting Block Information:-

(i) Num of addresses → $N = 2^{32-n}$

(ii) First address → (any address) AND (Network Mask)

(iii) Last " → (any address) OR [NOT (network mask)]

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Example 5.26 In the address $\underline{12}.23.24.78/8$

The network mask: $\underset{11111111}{255}.\underset{00000000}{0}.\underset{00000000}{0}.\underset{00000000}{0}$

The mask has eight 1
" " Twenty-four 0

∴ The prefix length is 8
" suffix " " 24

Example 5.27 One of the address in a block is 167.199.170.82/27. Find the number of addresses in the network, the first, the last add.

Solution:

The value of n = 27 → prefix length

The network mask has 27 1s and five 0s.

It is. $\underset{11111111}{255}.\underset{11111111}{255}.\underset{11111111}{255}.\underset{11100000}{224}$

The number of address in the network

$= 2^{32-n} = 2^{32-27} = 2^5 = 32$

Using AND operation.

Address:	167	199	170	82
Network mask:	255	255	255	224
First address:	167	199	170	64

→ next page

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$2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

Address (82): 0 + 1 + 0 + 1 + 0 0 1 0

Network mask (224): 1 1 1 0 0 0 0 0

First address: 0 1 0 0 0 0 0 0

To find last address. complement the network mask and then OR it

Address in binary	10100111	11000111	10101010	01010010
Network mask :	11111111	11111111	11111111	11100000
	10100111	11000111	10101010	01010010
complement of network mask :	00000000	00000000	00000000	00011111
Last add :	10100111	11000111	10101010	01011111
	167	199	170	95

Example 5.29 One of the addresses in a block is 110.23.120.14/20. Find number of address, the first address, last address.

Solution:

110.23.120.14/20

① Total number of address $= 2^{32-20} = 2^{12}$

$= 4096$

~~F~~ First address:

Address:	110	23	120	14
Network mask:	255	255	240	0
First add:	110	23	~~120~~ 112	0

Add:	0	1	1	1	1	0	0	0
mask:	1	1	1	1	0	0	0	0
first:	0	1	1	1	0	0	0	0

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